

INTERNSHIP PROJECT REPORT

Task 2: Intelligent FAQ Chatbot System

Intern Name: [Ayesha Aftab]

Internship Domain: Artificial Intelligence

Organization: CodeAlpha

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1. Project Overview

The objective of this project was to develop an intelligent FAQ Chatbot capable of understanding user queries and providing accurate responses from a predefined knowledge base. This project was completed as part of the **CodeAlpha AI Internship (Task 2)**. The system leverages Natural Language Processing (NLP) to bridge the gap between human language and machine understanding, ensuring a seamless user experience.

2. Technical Specifications

- **Programming Language:** Python 3.x
- **NLP Libraries:** NLTK (Natural Language Toolkit) for tokenization and lemmatization.
- **Mathematical Models:** TF-IDF (Term Frequency-Inverse Document Frequency) and Cosine Similarity.
- **User Interface:** CustomTkinter (Modern Dark-Themed Desktop GUI).
- **Text Processing:** String manipulation and punctuation filtering.

3. Key Features

- **Intent Matching:** Uses Cosine Similarity to find the most relevant answer, even if the user's phrasing is slightly different from the database.
- **Comprehensive Knowledge Base:** Contains 50+ detailed questions and answers regarding CodeAlpha's internship policies, perks, and submission guidelines.
- **Professional UI/UX:** A "Messenger-style" layout featuring a sidebar for branding, a scrollable chat area, and rounded input fields for a modern aesthetic.
- **Automated Preprocessing:** Automatically cleans and normalizes user input to improve the accuracy of the matching algorithm.

4. Methodology

The development process followed a structured engineering approach:

1. **Data Curation:** Collected and structured FAQ data into a Python dictionary for fast retrieval.
2. **Natural Language Processing:** Implemented a lemmatization pipeline using NLTK to reduce words to their root form (e.g., "running" to "run"), improving match consistency.
3. **Vectorization:** Applied TfidfVectorizer to convert text into numerical vectors based on word importance.
4. **Similarity Analysis:** Integrated cosine_similarity to calculate the mathematical distance between the user's vector and the FAQ vectors to identify the best response.
5. **GUI Construction:** Designed the interface using CustomTkinter, implementing a grid-based layout for responsiveness and a dark-tech theme for a professional look.

5. Results & Conclusion

The **CodeAlpha AI Assistant** effectively handles a wide range of internship-related queries with high precision. By combining traditional NLP techniques with a modern GUI, the system demonstrates the practical utility of AI in automating customer support and information retrieval. This project highlights the effectiveness of using similarity-based matching for specialized datasets.